Module 2 National Academic Standards (including Oklahoma State Standards):

National Academic Standards Next Generation Science Standards Common Core Standards for ELA and Math		
Science	**Science: SEP/DCI/CCC**	**ELA**
4.Earth's Systems: Processes that Shape the Earth **4-ESS3-2**. **5.Earth's Systems** **5-ESS2-1.** **5-ESS3-1.** **MS.Earth's Systems** **MS-ESS2-1.** **MS-ESS2-4.** **MS.Human Impacts** **MS-ESS3-4.** **HS.Human Sustainability** **HS-ESS3-1.** **HS-ESS3-4.** **HS-ETS1-2.**	**Science & Engineering Practice** ● Asking questions and defining problems ● Planning and carrying out investigations ● Analyzing and interpreting data ● Constructing explanations and designing solutions ● Developing and using models ● Obtaining, evaluating and communicating information ● Engaging in argument from evidence **Disciplinary Core Idea** ● ESS2.A: Earth materials and systems ● ESS2.C: The roles of water in Earth's surface processes ● ESS3.A: Natural resources ● ESS3.C: Human impacts on Earth systems ● ETS1.B. Developing possible solutions **Crosscutting Concepts** ● Patterns ● Cause and effect ● Scale, proportion, and quantity ● Scientific knowledge assumes an order and consistency in natural systems ● Systems and system models ● Energy and matter stability and change ● Influence of science, engineering, and technology on society and the natural world	**Writing Standards K–5 (Grades 4 & 5)** ***Text Types and Purposes*** **Sub-standard 2.a-e** ***Production and Distribution of Writing*** **Sub-standard 4** ***Research to Build and Present Knowledge*** **Sub-standards 7-9** **Speaking and Listening Standards K–5 (Grades 4 & 5)** ***Comprehension and Collaboration*** **Sub-standard 1.a-d** ***Presentation of Knowledge and Ideas*** **Sub-standard 4-6** **Writing Standards 6–12** ***Text Types and Purposes*** **Sub-standard 1.a-e.** **Sub-standard 2.a-f.** **Writing Standards 6–12** ***Production and Distribution of Writing*** **Sub-standards 4 and 6** ***Research to Build and Present Knowledge*** **Sub-standard 7** **Speaking and Listening Standards 6–12** ***Comprehension and Collaboration*** **Sub-standard 1.a-d** ***Presentation of Knowledge and Ideas*** **Sub-standards 4-6**

	• Science addresses questions about the natural and material world	

Oklahoma Academic Standards		
Science	**Science: SEP/DCI/CCC**	**ELA**
Earth's Place in the Universe (ESS1) **6.ESS2.1** **Earth and Human Activity (ESS3)** **6.ESS3.1** **Earth's Systems (ESS2)** **6.ESS2.4** **Earth and Human Activity (ESS3)** **7.ESS3.3** **7.ESS3.4** **HS Earth and Space Science (ES)** **Earth Systems (ESS2)** **ES.ESS2.5** **Earth and Human Activities (ESS3)** **ES.ESS3.1** **HS Environmental Science (EN)** **Earth Systems (ESS2)**	**Science & Engineering Practice** • Constructing explanations • Planning and carrying out/conducting investigations • Developing and using models • Obtaining, evaluating, and communicating information • Constructing explanations **Disciplinary Core Idea** • Earth's major systems are the geosphere, hydrosphere, atmosphere, and biosphere. • Human activities in agriculture, industry, and everyday life have had major effects on the land, vegetation, streams, ocean, air, and even outer space. But individuals and communities are doing things to help protect Earth's resources and environments. • Typically, as human populations and per-capita consumption of natural resources increase, so do the negative impacts on Earth unless the activities and technologies involved are engineered otherwise. • The abundance of liquid water on Earth's surface and its unique combination of physical and chemical properties are central to the planet's dynamics. These properties include water's exceptional capacity to absorb, store, and release large amounts of energy; transmit sunlight; expand upon freezing; dissolve and transport materials; and lower the viscosities and melting points of rocks. • Resource availability has guided the development of human society. **Crosscutting Concepts** • Patterns • Cause and effect • Scale, proportion, and quantity	**Standard 1: Listening and Speaking** ***All listed sub-standards*** **(Grades 4-12)** **Standard 3: Critical Reading and Writing** ***Reading sub-standards 6 through the last sub-standard*** **(Grades 4-8)** ***Reading sub-standards 5 through the last sub-standard*** **(Grades 9-12)** ***Writing sub-standard 2 through the last sub-standard*** **(Grades 4-12)** **Standard 6: Research** ***All listed sub-standards*** **(Grades 4-12)** **Standard 7: Multimodal Literacies**

EN.ESS2.5 **Earth and Human Activities (ESS3)** **EN.ESS3.1**	• Systems and system models • Influence of science, engineering, and technology on society and the natural world • Science addresses questions about the natural and material world	***All listed sub-standards*** **(Grades 4-12)**

Science: SEP (Science & Engineering Practice) DCI (Disciplinary Core Idea)
CCC (Crosscutting Concepts)